



XJTLU Entrepreneur College (Taicang)

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School Title	XJTLU Entrepreneur College (Taicang)
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Xi'an Jiaotong-Liverpool University

西交利物浦大學

Section 1 — Who You Tested With & How

Your participants

We tested our driving safety assistant with a diverse range of 8 participants, focusing on elderly drivers, young commuters, and high-frequency long-distance drivers.

Participant	Relevant background / why they were chosen	How you recruited them	Session format (in-person / online / etc.)
P1: Yang' mother	Who's father is a driver over 70; represents elderly users with resistance to complex tech	Family connection	In-person interview
P2: LJZ	University student; represents young cross-city night commuters.	Schoolmate	In-person interview
P3: Wang Guohua	41-year-old; professional commuter driving long distances at night	Alumni resources	Online (WeChat)
P4: Li Fu	65-year-old; drives a 15-year-old vehicle without modern safety tech	Community resources	Online (WeChat)
P5: Mr. Li	Adult male driver; high standards for cabin organization.	Schoolmate	In-person interview
P6: Aunt Liu	Elderly driver of short stature; shares vehicle with a taller	Community resources	Online (WeChat)

	son		
P7: Teacher Liu	School teacher; needs remote monitoring for elderly mother	Community resources	Online (WeChat)

What you asked them to do

- **Initial Interviews:** Discussed driving habits, fatigue experiences, and barriers to using existing navigation tools.
- **Prototype Testing:** Users integrated the hardware box onto their dashboards and activated it using the blue "start" button.
- **Hazard Identification:** When alarms were triggered (fatigue or speeding), users were asked to identify the hazard and respond to the feedback.
- **App Interaction:** Later participants used the companion app to calibrate sensor sensitivity and adjust alarm volumes.
- **Duration:** Sessions lasted approximately 15–30 minutes.

How you captured what happened

We used a combination of interview recordings, detailed session notes, and screenshots of WeChat follow-up conversations. Online sessions were documented via screen captures of text and voice messages.

Section 2 — What You Found

Finding 1: Feedback Ambiguity and Distraction

During the night test with P3 (Wang Guo-hua), it was found that the simple "buzzer OLED small screen" solution had safety hazards. Users have reported that it is difficult to distinguish whether it is a "fatigue warning" or a "speeding warning" just by sound while driving. They

have to look away at the text on the screen, which instead increases the risk of distraction when driving at high speed (Session Note 3).

Finding 2: Hardware Obstruction and Size Issues

Tests on elderly users (P4) revealed that the text on the original OLED screen was too small and in English, causing them to feel flustered and unable to understand the instructions when the alarm was triggered. Meanwhile, the high-brightness red LED is too dazzling in the night environment and can interfere with the driver's observation of the road conditions (Session Note 4).

Finding 3: Fixed Sensing Thresholds Cause False Alarms

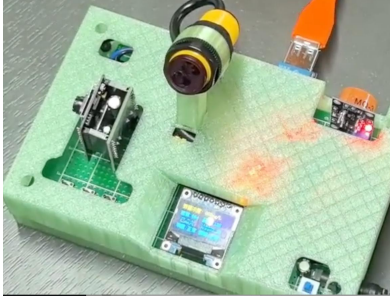
Mr. Li pointed out during the actual vehicle installation that the experimental prototype based on the breadboard was too large in size, blocking the car's central control screen and some air conditioning control buttons. The user explicitly stated that they would not install and use the device for a long time if it interfered with the operation of the vehicle's native functions (Session Note 6).

Finding 4: UI Accessibility for Elderly Users

This is the most serious logical error found in the test. When testing the P7 (Aunt Liu, who is smaller in stature), as she needed to move the seat forward significantly to step on the pedal, the infrared sensor would continuously determine it as "drowsiness with the head down" due to the overly close distance and give an incorrect alarm. This "one-size-fits-all" fixed trigger distance cannot adapt to users of different heights and driving habits (Session Note 7).

Section 3 — What You Changed

What finding triggered this change	What you changed (before → after)	Evidence that the change helped
Ambiguous feedback	Before: Single buzzer/OLED. After: Multi-modal "Screen + Voice"	In LJZ's interview (session 5) video, it successfully responded to specific voice prompts: "Please pay attention to rest".

	system with clear hazard broadcasts.	Video Link: Video Evidence 5
Size obstruction	Before: Large pegboard prototype. After: Compact PCB and 3D-printed enclosure.	Planned miniaturization to prevent blocking the dashboard. Picture: 
False alarms (Height)	Before: Fixed trigger distance. After: IoT App allowing custom sensitivity and distance calibration.	P7 reported the app successfully resolved false alarms for her shorter mother. Relevant Session note: Session Note 8
UI/Night Glare	Before: Small English text; bright red LED. After: Large Chinese text; added light diffuser cap.	Improved readability and reduced nighttime distraction for elderly drivers. Picture: 

Most significant change — explained

The most critical iteration was the transition to an **IoT-based architecture with a companion app**. This solved the fundamental "one-size-fits-all" failure of the initial hardware. By moving complex settings to the app, we allowed for **multi-driver profiles**.

This ensured that users of different statures, like Aunt Liu and her son, could use the same device without false alarms, making the product viable for shared family vehicles.

What you chose not to change — and why

We decided not to remove the alcohol monitoring module entirely, even though P5 found it redundant. Instead, we planned to make it a **modular optional component**. This preserves the system's comprehensive safety value for target groups who do require it while allowing for the miniaturization requested by users like P5.

Section 4 — Limitations

- **Sample size:** How many people did you test with? What kinds of users could you not reach?

While we reached 7 participants across key demographics, we still lack testing with high-capacity professional truck drivers whose fatigue patterns may differ from commuters.

- **Testing conditions:** Were sessions realistic? What was different from real-world use?

Many sessions relied on simulated fatigue or stationary cabin environments. Real-world high-speed emergency reactions to voice prompts remain to be fully validated.

- **What you could not test:** Were there features or scenarios you were not able to test?

Up to now, we still haven't been able to find groups such as leaders of large truck or taxi companies for testing. Therefore, this product is still limited to family-oriented usage scenarios and has not been expanded to a larger market

- **Long-term Effects:**

We have not yet evaluated "alarm fatigue," where a user might begin to ignore or disable the system after weeks of habitual use.

Section 5 — Evidence Appendix

Reference	What it contains	Link
Session Note 1	Session Note 1: Elderly Driver Interview	https://xjtlums-my.sharepoint.com/:w:/g/personal/zhiqin_yang23_student_xjtl_u_edu_cn/IQA8wUyJr31mQJuvFDzLCOosARVRp_2QskSKiKQYS8QAK-A?e=rnPQIZ
Session Note 2	Session Note 2: Young Commuter Interview	https://xjtlums-my.sharepoint.com/:w:/g/personal/zhiqin_yang23_student_xjtl_u_edu_cn/IQCNapSd3kwAS5ZPG6H146CRAdD7Qff_eQPnHj6T4vsI9QM?e=ol70Sv
Session Note 3	Session Note 3: Multi-modal Alarm Interaction & Information Clarity	https://xjtlums-my.sharepoint.com/:w:/g/personal/zhiqin_yang23_student_xjtl_u_edu_cn/IQAWZSLK4p3MS6cqERaVTgtFAblJxEq7y3ouLgPat0AF0?e=6o3OBO
Session Note 4	Session Note 4: In-Cabin Hardware Interaction & Alarm Experience	https://xjtlums-my.sharepoint.com/:w:/g/personal/zhiqin_yang23_student_xjtl_u_edu_cn/IQC8UCgZxegDS7vwSSCwkyJxATh4tUgVi8rJ_0zyrAMyjs?e=ebCrFh
Session Note 5	Session Note 5: Trigger Logic Calibration	https://xjtlums-my.sharepoint.com/:w:/g/personal/zhiqin_yang23_student_xjtl_u_edu_cn/IQBkwJL3mJLcS4ZHewR17JP2AYa7WG9LAESC0FEHrE529cE?e=zPfGJg
Session Note 6	Session Note 6: Device shape and space occupation	https://xjtlums-my.sharepoint.com/:w:/g/personal/zhiqin_yang23_student_xjtl_u_edu_cn/IQCeIXdn5RAKTZqT9WBAhSRsAZlg3WA0V7Lp8NvB1ulpjww?e=LWC9ks

Session Note 7	Session Note 7: Sensor Calibration & UI Usability for Cross-generational Drivers	https://xjtlums-my.sharepoint.com/:w:/g/personal/zhiqin_yang23_student_xjtl_u_edu_cn/IQDyKOD9_k_DSJztXTacKe20AXy26SUseWIn6AD8lmGbJZI?e=cOjNxt
Session Note 8	Session Note 8: Research on Custom IoT Configuration and Remote Safety Monitoring Requirements	https://xjtlums-my.sharepoint.com/:w:/g/personal/zhiqin_yang23_student_xjtl_u_edu_cn/IQCO958VF5P3Rbd94RvquEXkAbMHejHCXJbJJs7j5VIUUOA?e=SCMa7m
Video Evidence 2	LJZ Problem Validation Interview Recording	https://xjtlums-my.sharepoint.com/:v:/g/personal/zhiqin_yang23_student_xjtl_u_edu_cn/IQDjVu0eveNFRLeuYAhVbrTCAZKggXtHhkQKg9Z2cD8wRvk?nav=eyJyZWZlcnJhbEluZm8iOjnsicmVmZXJyYWxBcHAiOiJPbmVEcmI2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlc nJhbFZpZXciOiJNeUZpbGVzTGlua0Nvc HkifX0&e=f2kaaR
Video Evidence 5	LJZ Early Prototype Interview Recording	https://xjtlums-my.sharepoint.com/:v:/g/personal/zhiqin_yang23_student_xjtl_u_edu_cn/IQDAZk6wFKrbRpFjAHZi4dieASuxRXL_n2X7u_JV Kr45L4g?nav=eyJyZWZlcnJhbEluZm8iOjnsicmVmZXJyYWxBcHAiOiJPbmVEcmI2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFs
Video Evidence 6	Mr. Li Early Prototype Interview Recording	https://xjtlums-my.sharepoint.com/:v:/g/personal/zhiqin_yang23_student_xjtl_u_edu_cn/IQAOdDFmQz9JTLidNQjIXIbAAQHESMN9zBQntKhrsSUqK24?nav=eyJyZWZlcnJhbEluZm8iOjnsicmVmZXJyYWxBcHAiOiJPbmVEcmI2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlc nJhbFZpZXciOiJNeUZpbGVzTGlua0Nvc HkifX0&e=l4abhr

Recording Evidence 1	Interview with Yang's mother	https://xjtlums-my.sharepoint.com/:u:/g/personal/zhiqin_yang23_student_xjtlu_edu_cn/IQCcLGIQchffQYEvuzV5eTZhAfbOcsjFLCJSN28IkUveW9I?nav=eyJyZWZlcnJhbEluZm8iOnsiOiJPbmVEcmI2ZUZvckJlc2luZXNzIiwicmVmZXJyYWxBcHBqbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkiX0&e=TlQxq3
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